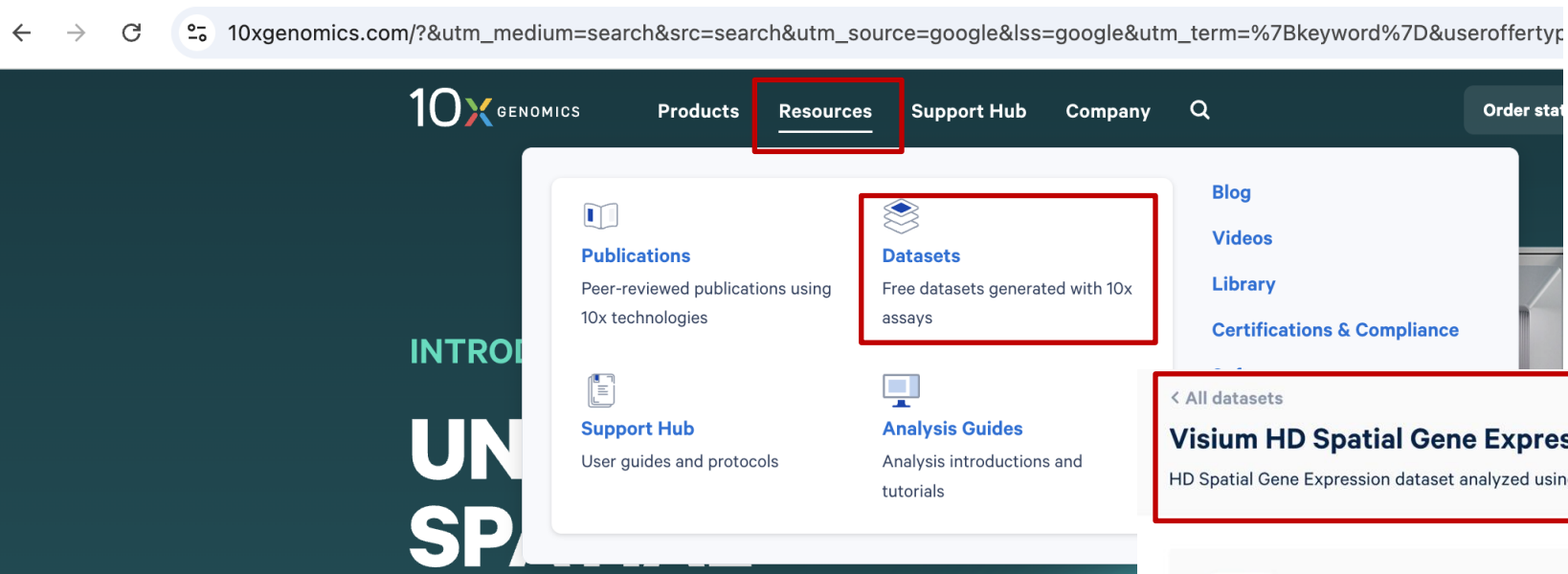


CRI Bioinformatics Bootcamp 2026

Day 6: Introduction to Spatial Transcriptomics

Nataly Naser Al Deen, PhD | Memorial Sloan Kettering Cancer Center

Sample Pre-processing: Visium HD



Visium HD Spatial Gene Expression Library, Human Breast Cancer, IF (FFPE) - 10X genomics

<https://www.10xgenomics.com/datasets/visium-hd-cytassist-gene-expression-libraries-human-breast-cancer-ffpe-if>

< All datasets

Visium HD Spatial Gene Expression Library, Human Breast Cancer, IF (FFPE)

HD Spatial Gene Expression dataset analyzed using Space Ranger 3.1.2

Mapped
95.7%

Assess data quality
View summary metrics to assess data quality and more.

View summary

Visualize and explore data
Discover differentially expressed genes, visualize your favorite genes, and explore your data with our visualization software.

Explore data

Dataset overview

Output and supplemental files

Input files

[Learn about Visium Analysis](#)

Raw output

Files

Learn more about file formats

Batch download

Output files	File type	Size	md5sum
Web Summary HTML	HTML	11.2 MB	887a4f53906216c699d6bfa20cd5d53e
Loupe file	CLOUPE	3.92 GB	20b2458a29eeda82331f9a5b33e37a9a
Feature slice H5	H5	494 MB	08b699e1b08cf18dca80a781325eaa05
Metrics summary CSV	CSV	1.82 kB	fe61e18ff353bcf540dba7960242ba1a
Molecule info H5	H5	496 MB	f4344d40142452247859a6ff7cdbe4c
Spatial outputs	GZ	66.2 MB	af849436ba4615857be11365dc2aff3b
Binned outputs (all bin levels)	GZ	6.75 GB	9dce8b788df7eb9f3802c87c72eff3cd

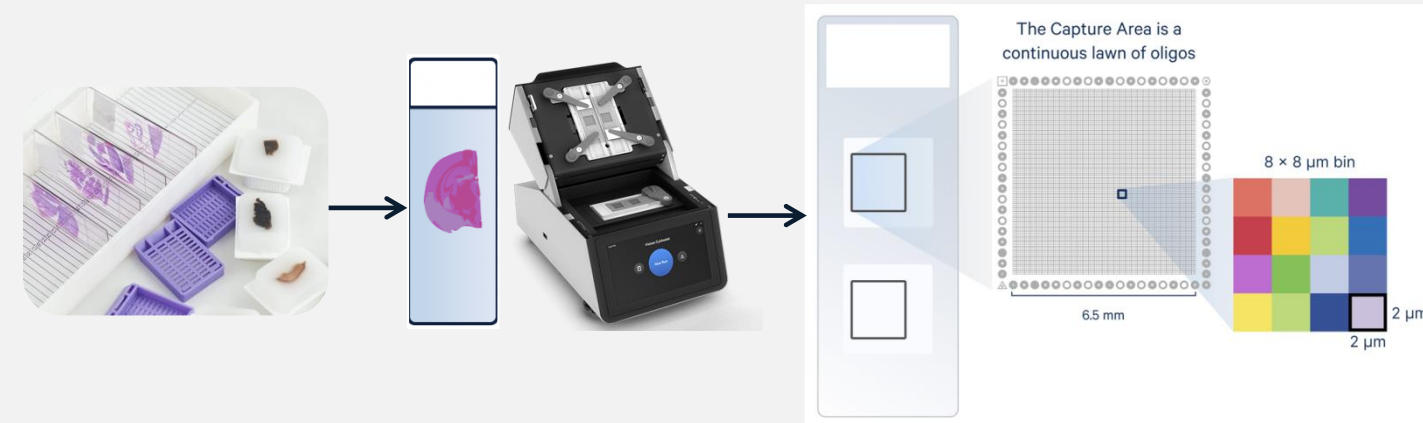
Visium Spatial Transcriptomics HD enables 18,000 whole transcriptome on 5µm sections of FFPE blocks or archived (un)stained slides with DV200>30% at a (2µmx2µm) resolution

Input: 5µm FFPE/OCT tissue (6.5x6.5mm)

Output: high resolution H&E (brightfield) or IF (Fluorescent) image (≈1GB) and FASTQ files

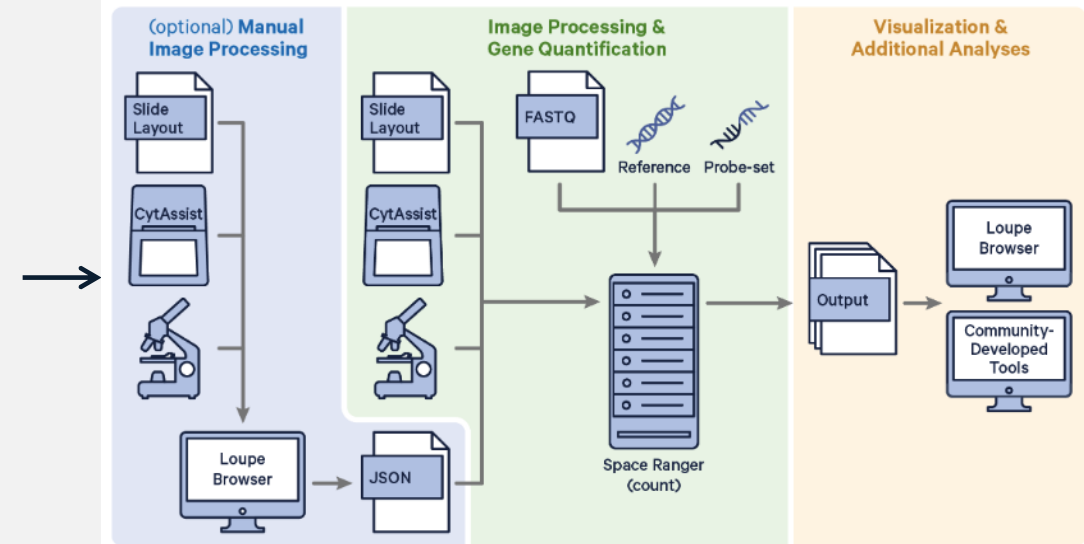
A

Visium HD experimental planning



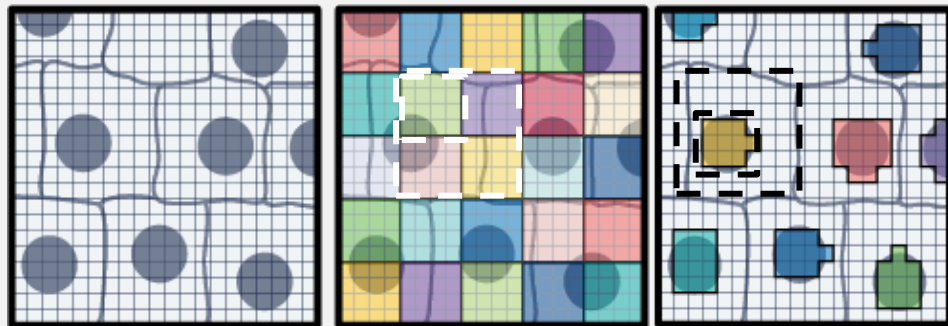
B

Space Ranger Alignment



C

Nuclei segmentation and custom binning



D

Third party tool analysis



Input files

Before running the pipeline, check that you have the following inputs prepared:

- The corresponding [CytAssist image](#) in `TIFF` format (`--cytaimage`)
- [Microscope image](#) (optional) in either `TIFF`, `QTIFF`, `BTF`, or `JPEG` format:
 - `--image` for a brightfield microscope image
 - `--darkimage` for a dark background fluorescence microscope image
 - `--colorizedimage` for a composite colored fluorescence microscope image
- [Slide parameters](#) specified by:
 - `--slide` & `--area` if `spaceranger` has access to internet (optional with CytAssist metadata)
 - `--slidefile`, `--slide` & `--area` if `spaceranger` has no access to the internet (the slide layout file must be [directly downloaded](#))
 - `--unknown-slide` if Visium slide details are unknown
- The reference transcriptome (`--transcriptome`). [Human and mouse references are available for download](#), or [build a custom reference](#)
- The [probe set CSV](#) (`--probe-set`). These files can be found in the `probe_sets` directory in the Space Ranger software package or [downloaded here](#)
- In most cases Space Ranger will perform automatic image to fiducial alignment and tissue detection. When automatic alignment fails, you must also run [manual alignment](#) in Loupe Browser and specify the alignment JSON file with the `--loupe-alignment` option.

SpaceRanger Input: high resolution H&E (brightfield) or IF (Fluorescent) image (≈1GB) and FASTQ files (sometimes json files)

SpaceRanger Outputs: can be in customized bins, multiple of 2, up to 100x100μm bin

☐ Show Owner/Mode ☐ Show Dotfiles Filter:

Showing 3 rows - 0 rows selected

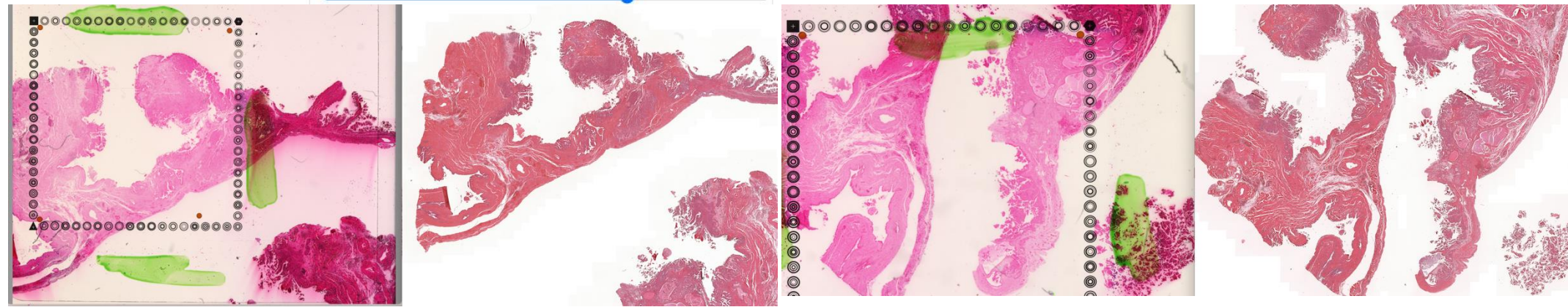
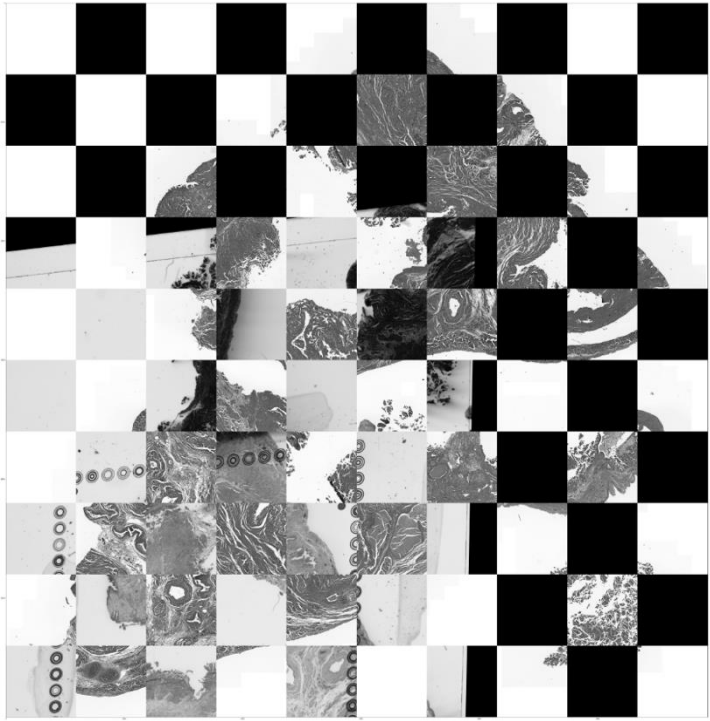
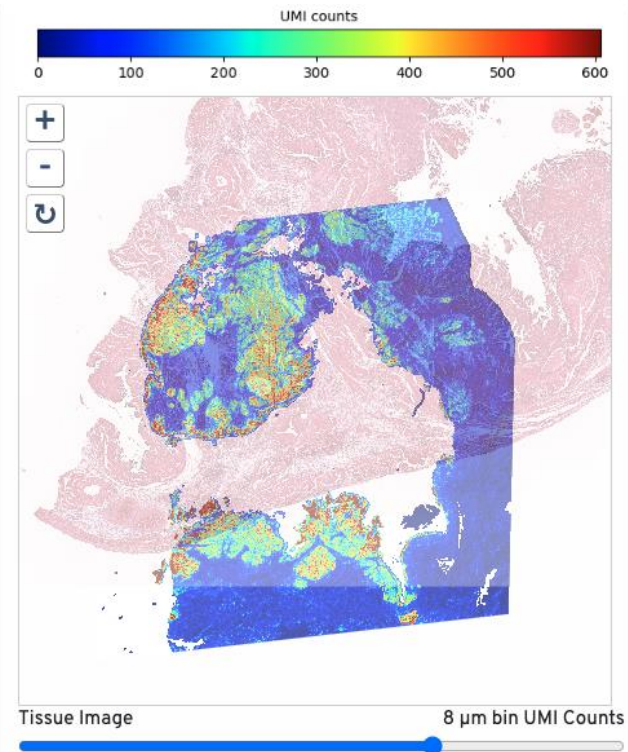
Type	Name	Size	Modified at
☐ 	square_016um	-	12/13/2024 2:02:00 AM
☐ 	square_008um	-	12/13/2024 2:01:51 AM
☐ 	square_002um	-	12/13/2024 2:01:26 AM

We choose the 8μmx8μm bin to achieve closer to a single cell and since it is the smallest bin that can produce a cLoupe Browser interactive file

Tissue Troubleshooting: First step is to always examine the outputs **web summary** for any errors, warnings, or tissue mis-alignment

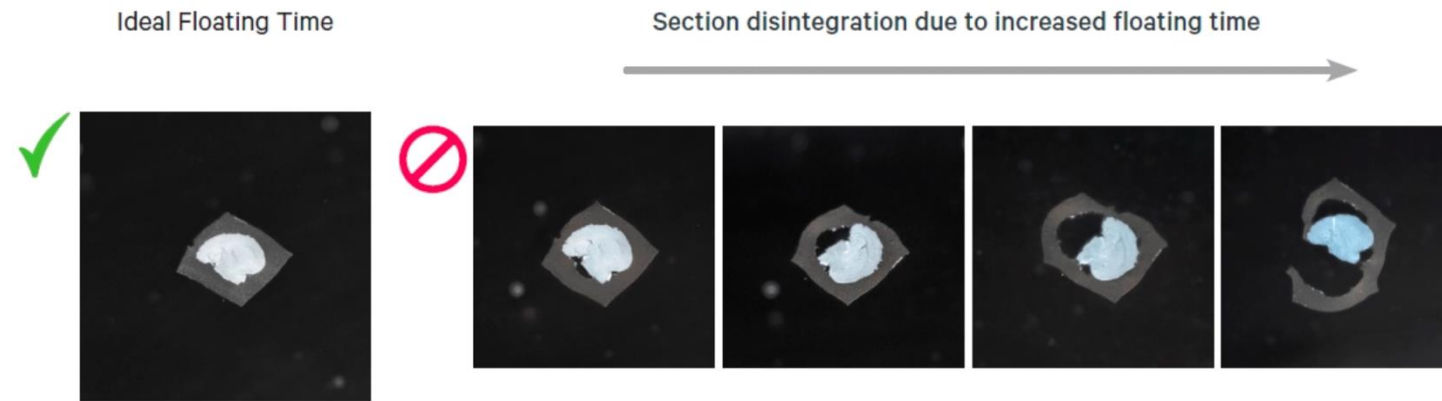
Reads Half-Mapped to Probe Set	0.2%
Reads Split-Mapped to Probe Set	0.1%

Sequencing ?	
Number of Reads	781,188,369
Valid Barcodes	95.6%
Valid UMIs	99.7%
Sequencing Saturation	90.4%
Q30 Bases in Barcode	98.1%
Q30 Bases in Probe Read	97.9%
Q30 Bases in UMI	97.0%
Fraction of Bins Under Tissue 8 μ m	59.1%
Fraction Reads in Squares Under Tissue	94.1%

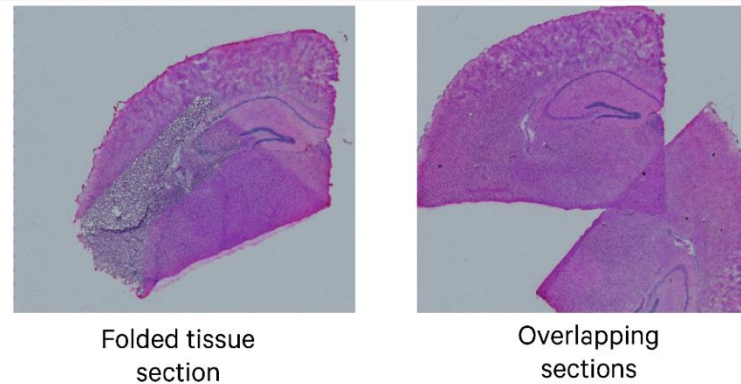


Troubleshooting

Ideal Floating Time Determination



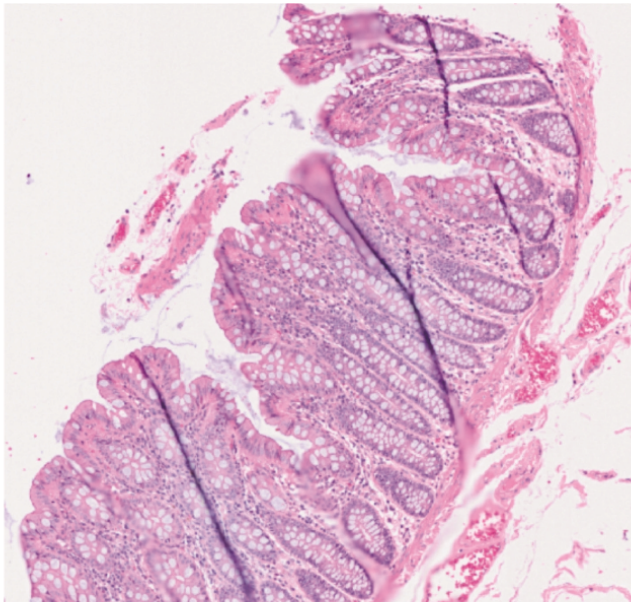
Incorrect Placement of Tissue Sections



Tissue Troubleshooting: Prior to the Visium HD CytAssist alignment by examining the H&E QC section

Common Artifacts that cause Detachment

Wrinkles



Causes

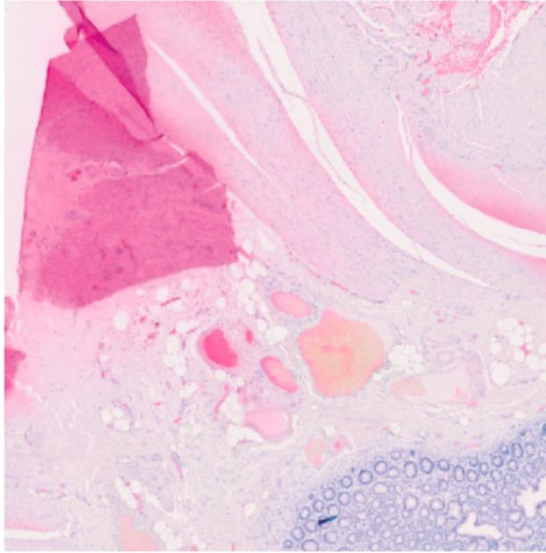
- Section compression (due to warm block or dull blade) during sectioning leads to wrinkle formation. These wrinkles become permanent when placed in the water bath.
- Accumulated wax or static electricity on microtome parts also contribute to section compression.

Troubleshooting

- Ensure that the block is well hydrated.
- Adjust temperature down and increase float time.
- Gently and gradually lay FFPE sections onto water bath surface, lengthwise.
- Utilize a new blade.
- Ensure microtome is cleaned with 100% ethanol to minimize static and section compression (bunching on blade).

Tissue Troubleshooting: Prior to the Visium HD CytAssist alignment by examining the H&E QC section

Folds



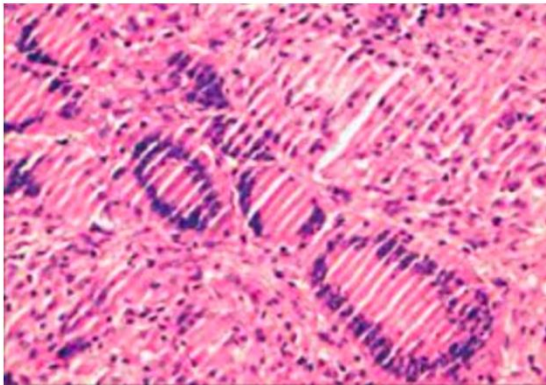
Causes

- Mostly happens when placing the section on the water bath especially when the section is uneven.
- If the fold is at the edge this most likely can happen during sectioning or mounting on the slide.

Troubleshooting

- Gently and gradually lay FFPE ribbons or sections onto water bath surface, lengthwise.
- If sections curl during sectioning, gently flatten them with a brush before floating.

Venetian Blinds or Shatter



Causes

- Parallel lines in the section mostly appear due to dry tissue because of under-hydration of the block in the ice bath.
- Less likely due to dull blade or loose parts of the microtome.

Troubleshooting

- Increase incubation time of the block in an ice bath.
- Tighten down components of microtome and make sure the blade is at a correct angle.

While samples should be carefully QC-ed before proceeding to the Visium HD protocol for any detachment, folding, and DV200, in some instances where no tissue block is available (depleted block), and tissues have already been sectioned and archived, correcting for these mis-alignment/detachment can be done after sequencing and before pre-processing using 10X Loupe Browser

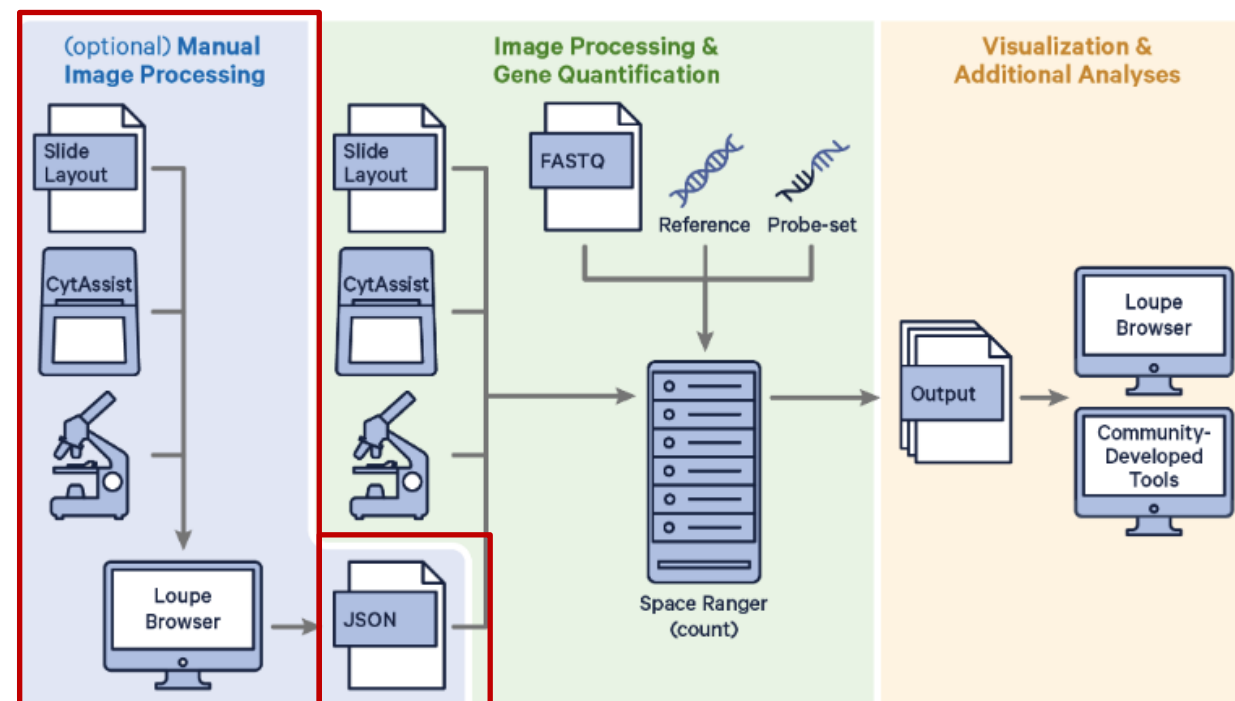
Tissue Troubleshooting: After the Visium HD Run

Few instances for manual fiducial alignment:

- when the tissue covers the fiducial frame (old Visium only, no issue with Visium HD)
- when there is any tissue detachment
- Sometimes with deeper sequencing, some empty bins might have bleeding over of signal from neighboring bins with tissue present

Visium HD Analysis with spaceranger count

Space Ranger v3.0 and later can analyze Visium HD Spatial Gene Expression datasets generated from formalin fixed paraffin embedded (FFPE) human or mouse samples and the Visium CytAssist instrument. The `spaceranger count` pipeline is necessary to analyze these data.

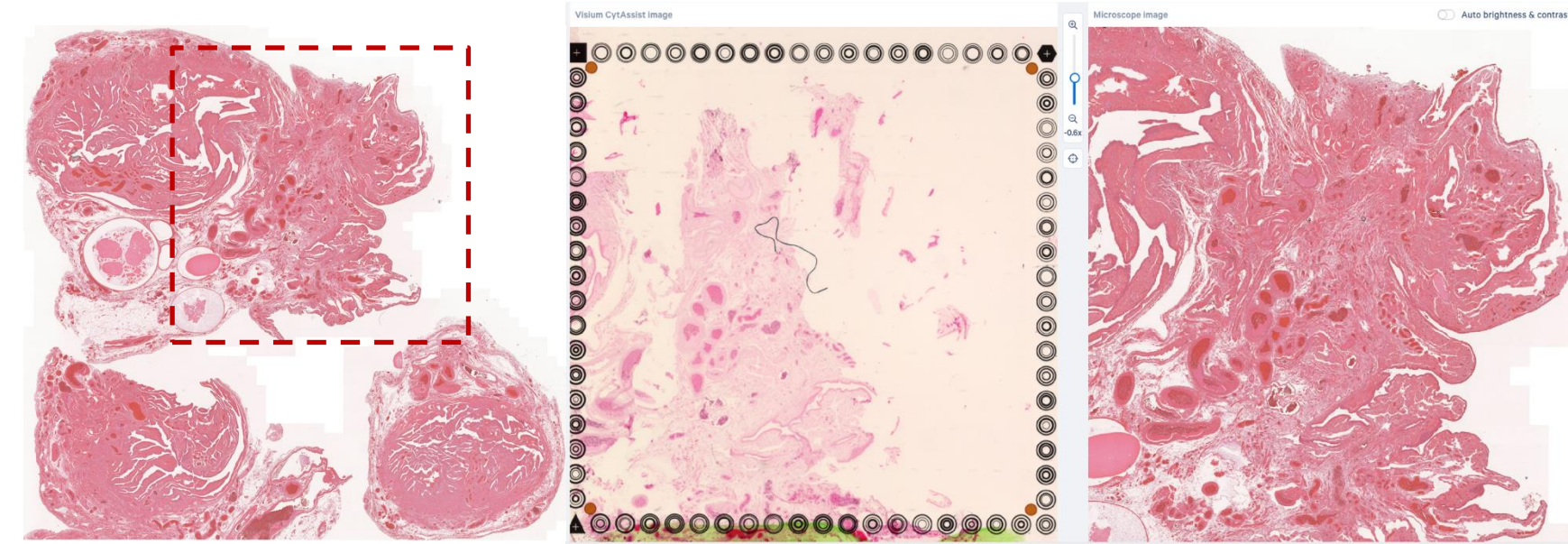


Input files

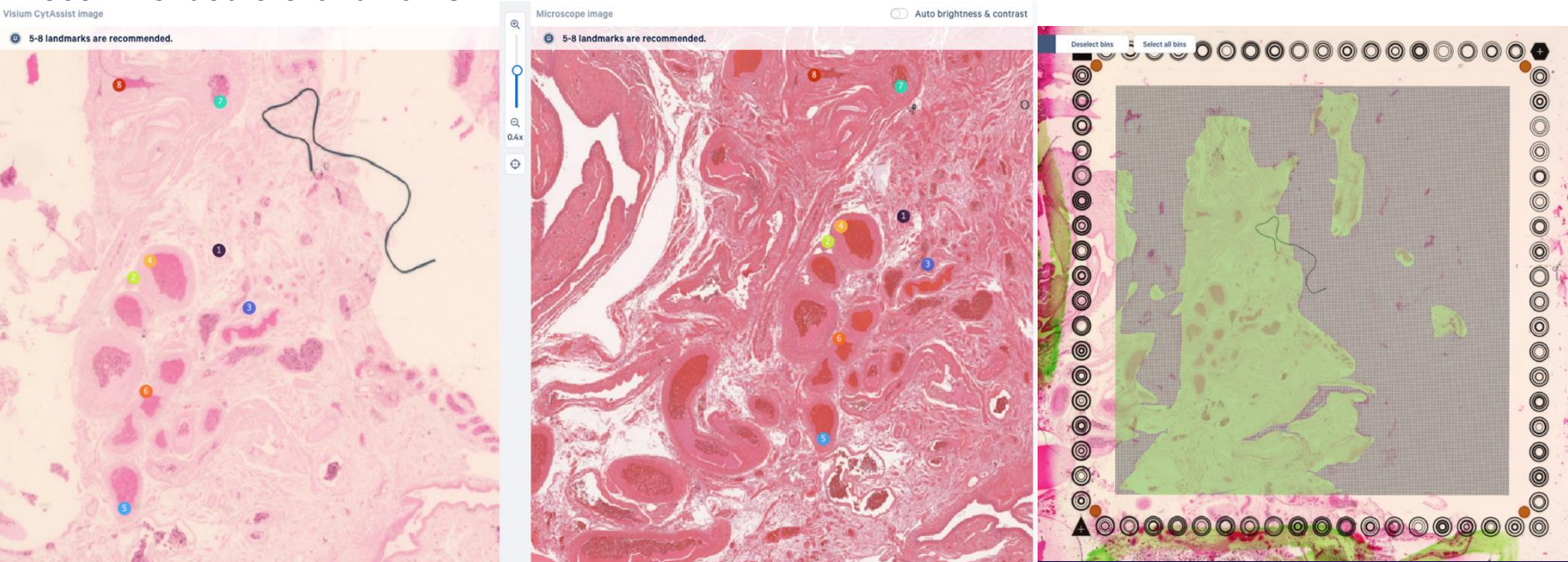
Before running the pipeline, check that you have the following inputs prepared:

- The corresponding **CytAssist image** in **TIFF** format (`--cytaimage`)
- Microscope image** (optional) in either **TIFF**, **QTIFF**, **BTF**, or **JPEG** format:
 - `--image` for a brightfield microscope image
 - `--darkimage` for a dark background fluorescence microscope image
 - `--colorizedimage` for a composite colored fluorescence microscope image

Tissue Troubleshooting After the Visium HD Run: Manual fiducial alignment when there is any tissue detachment

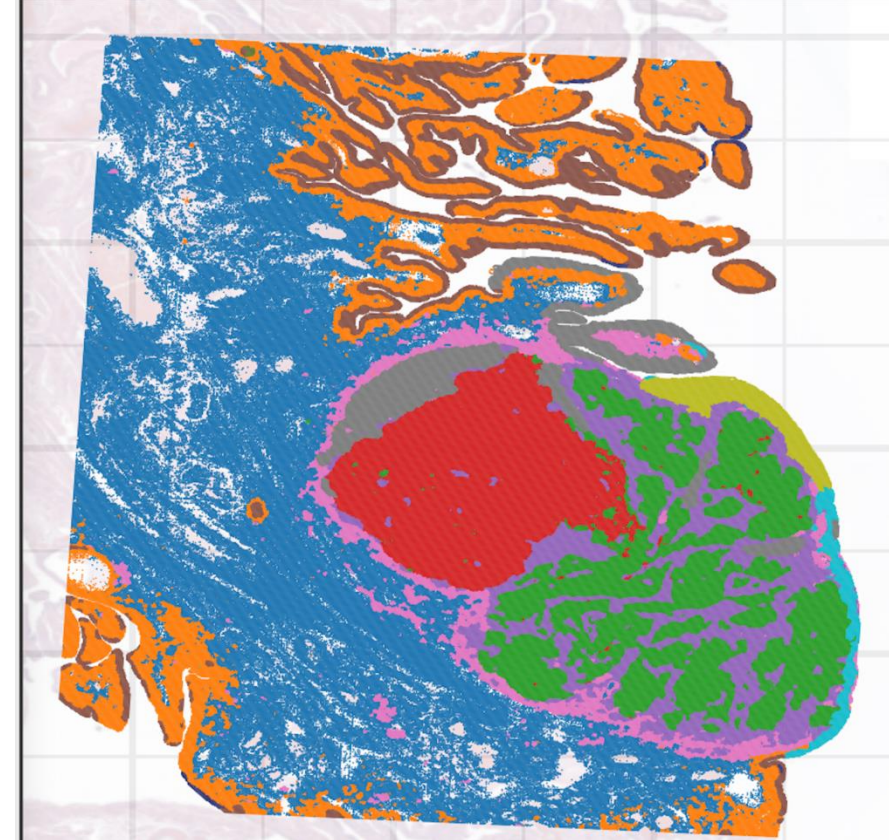
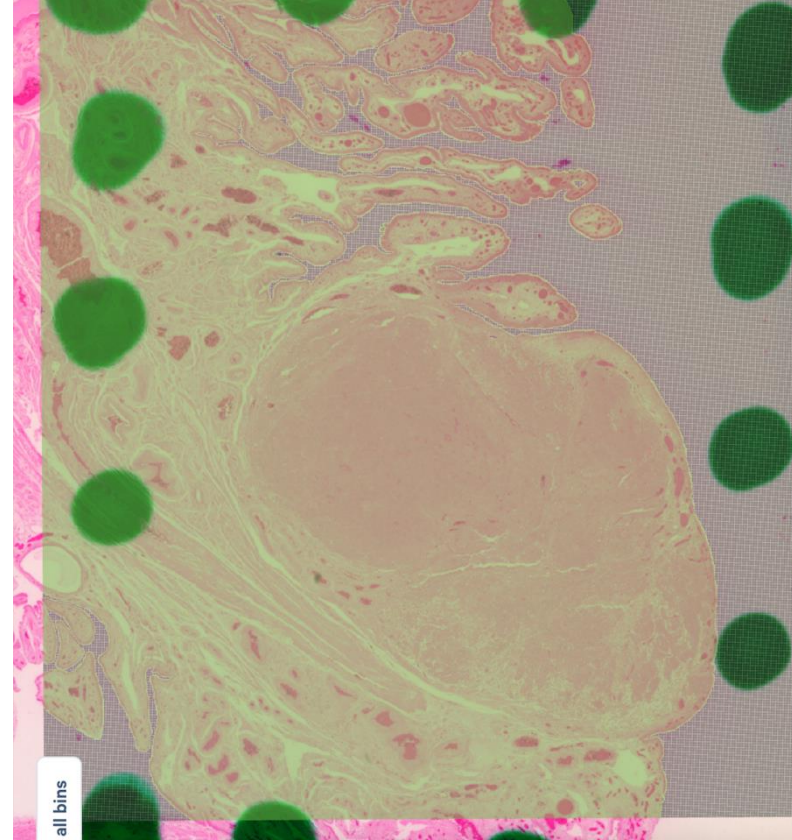
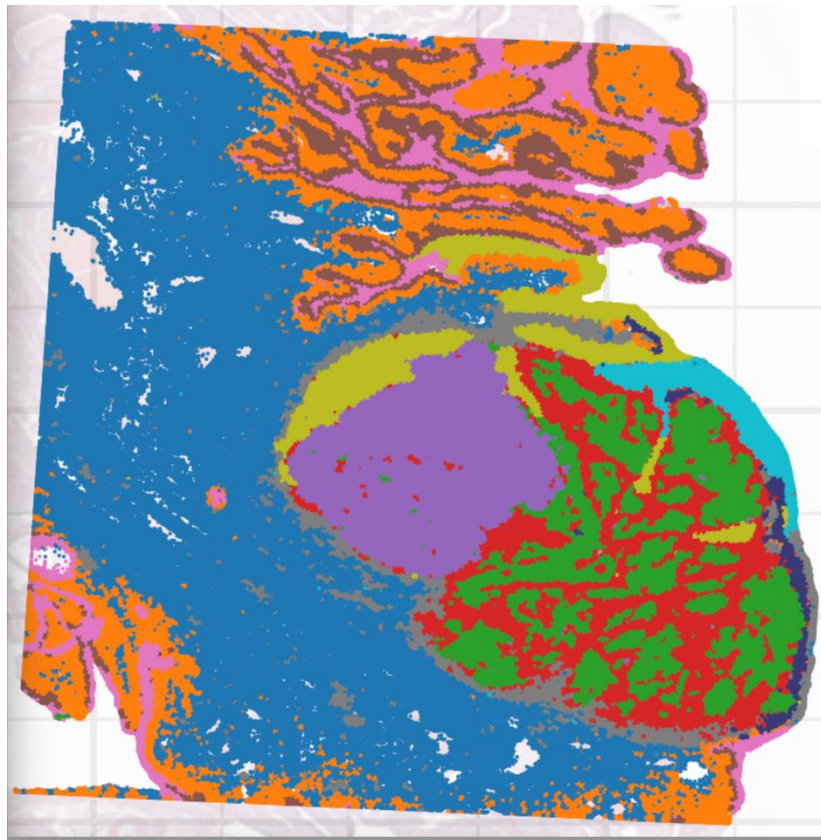


Recommended 5-8 landmarks



Recommended slides to avoid detachment: Schott Nexterion Slide (Schott, PN-1800434)

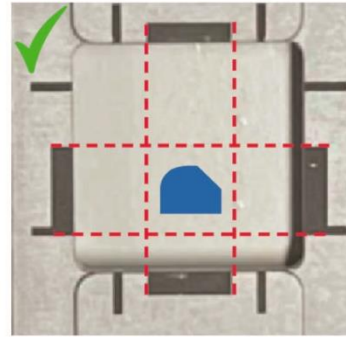
Tissue Troubleshooting After the Visium HD Run: With opting for deeper sequencing, some empty bins might have bleeding over of signal from neighboring bins with tissue present



The actual working area is **6x6mm** and not 6.5x6.5mm. Avoid placing your important ROI closer to the edges while aligning the H&E/IF slide to the CytAssist gasket

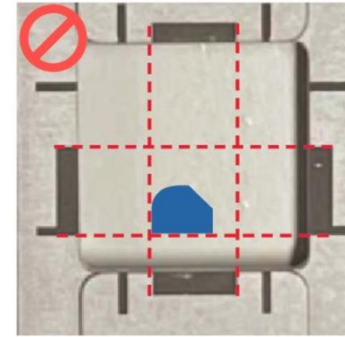
Correct

Extend lines from the alignment guides to create an imaginary square. Place the tissue on the center of that square.

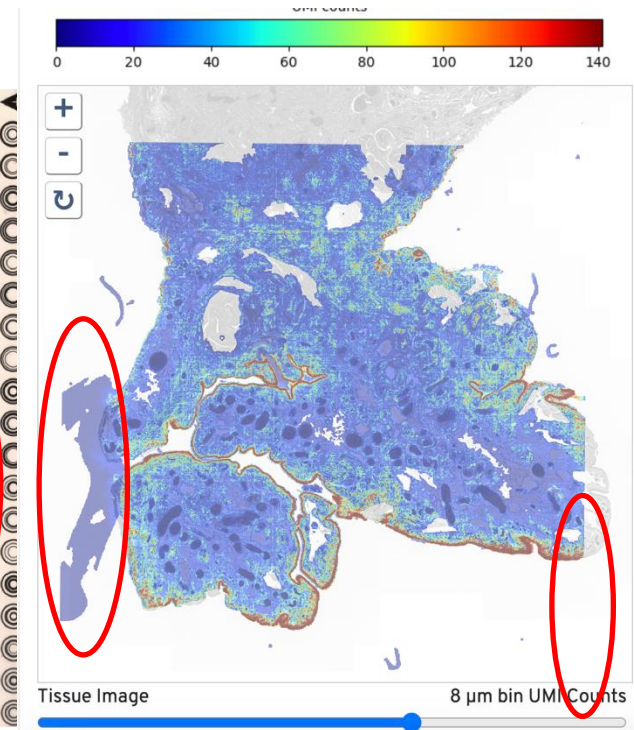
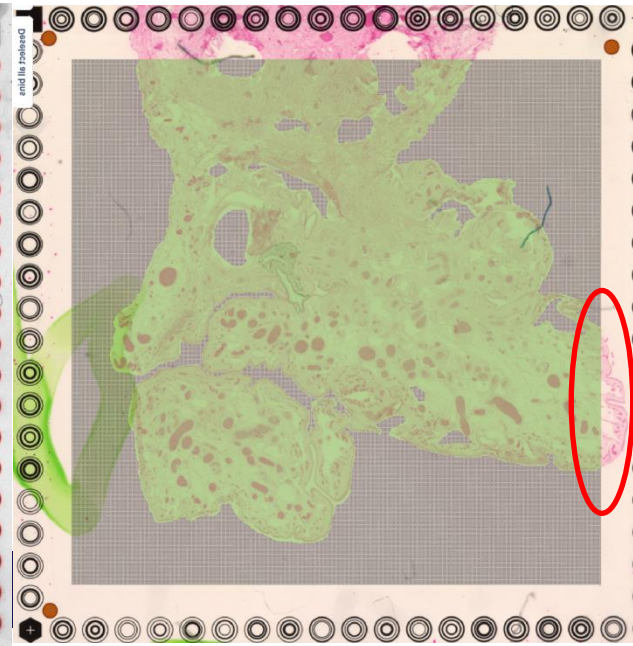
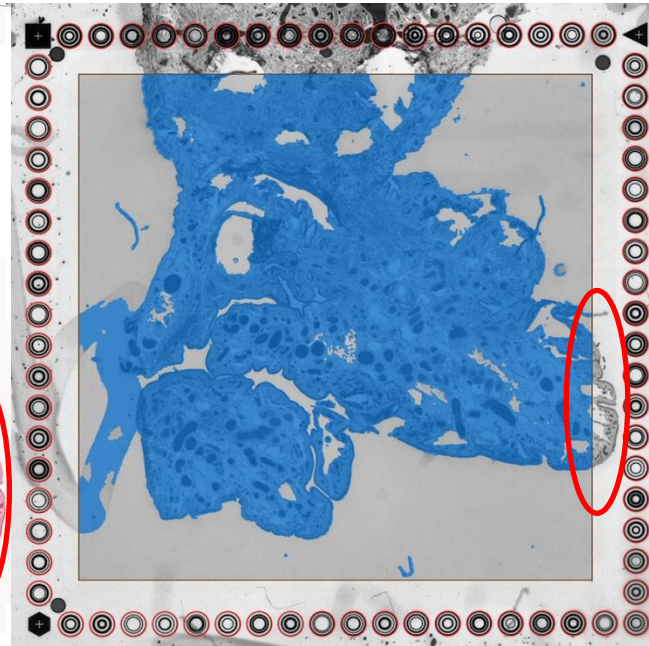
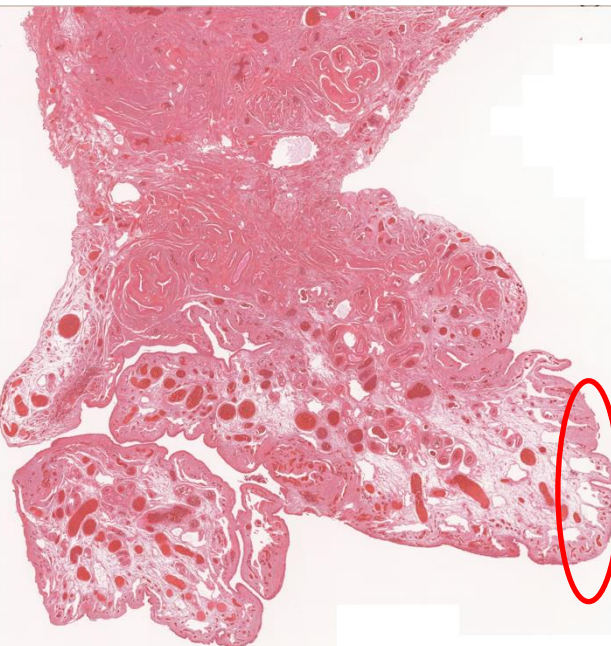


Incorrect

Tissue is within the alignment guides, but should be centered within the imaginary square, not aligned with the top or bottom line.

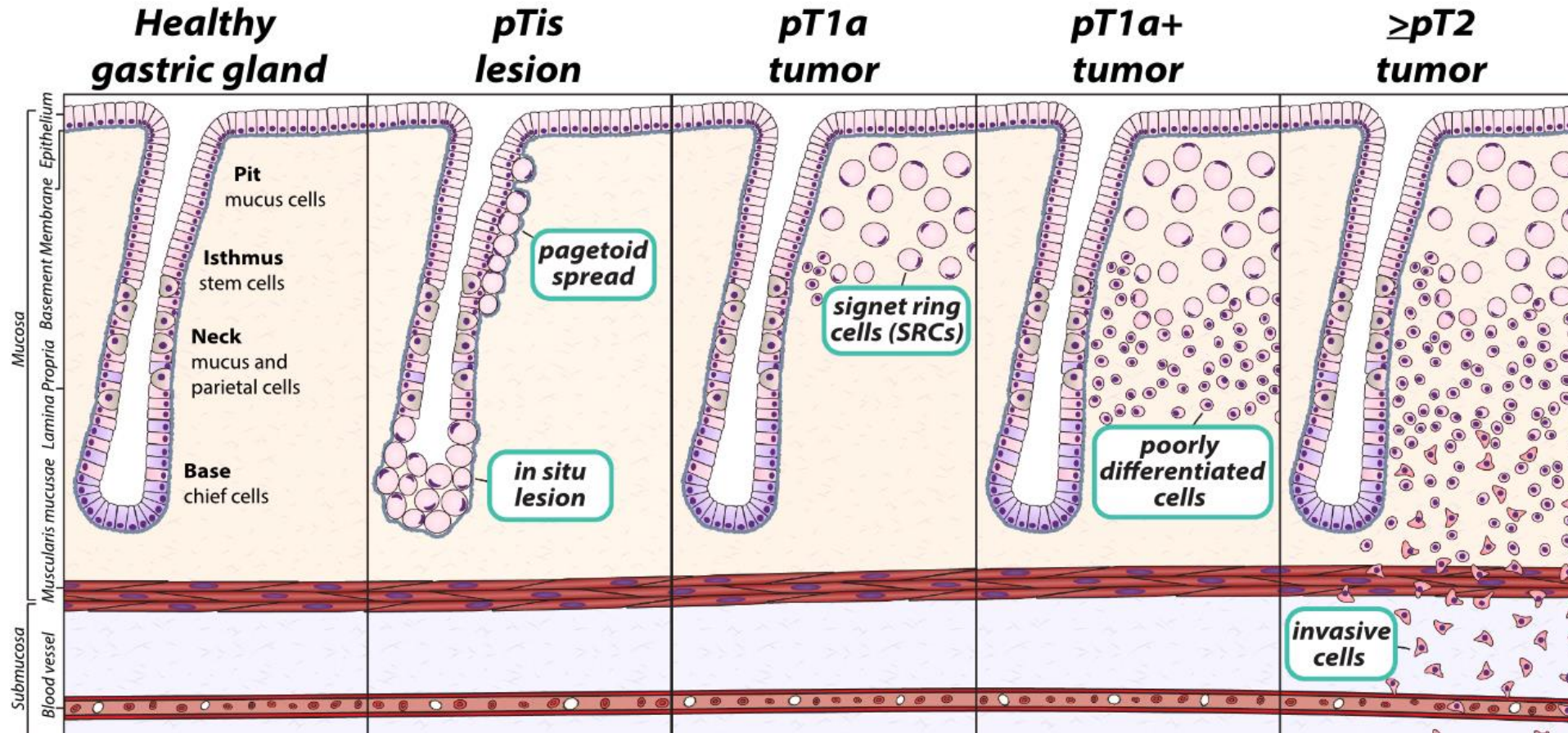


6.5 mm capture area used as an example



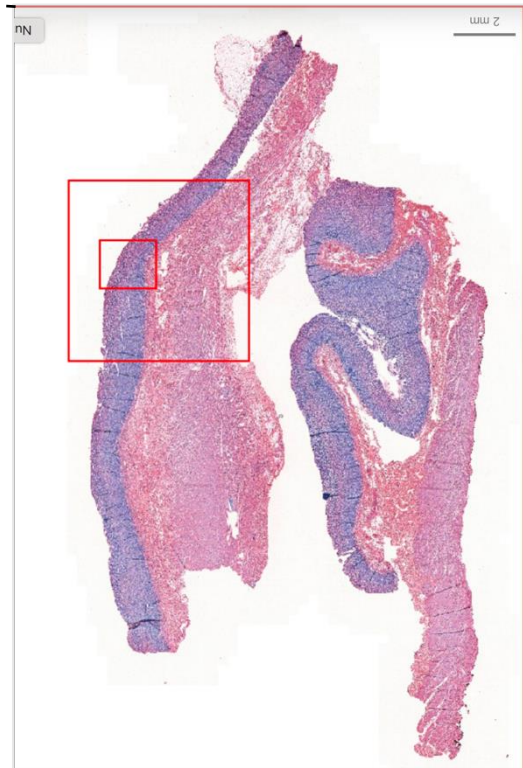
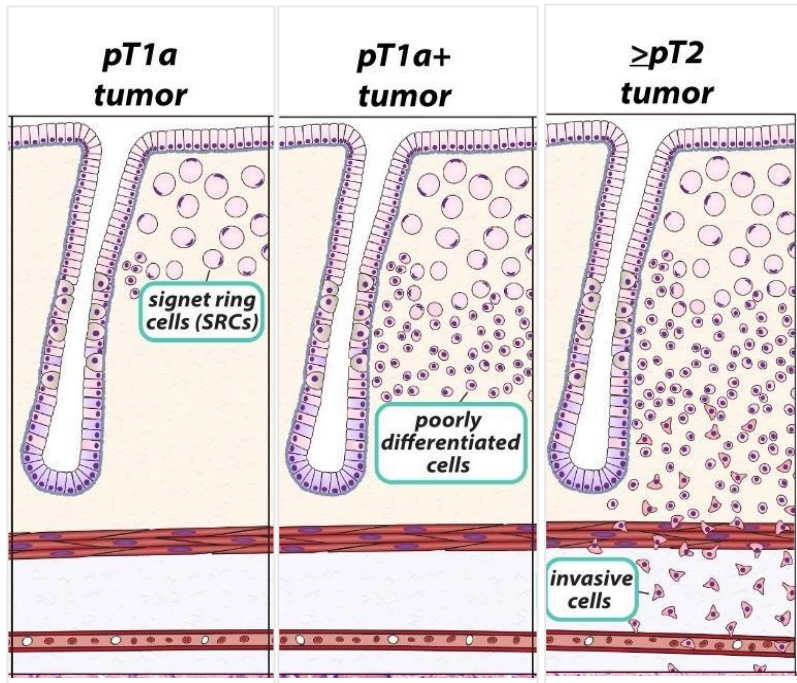
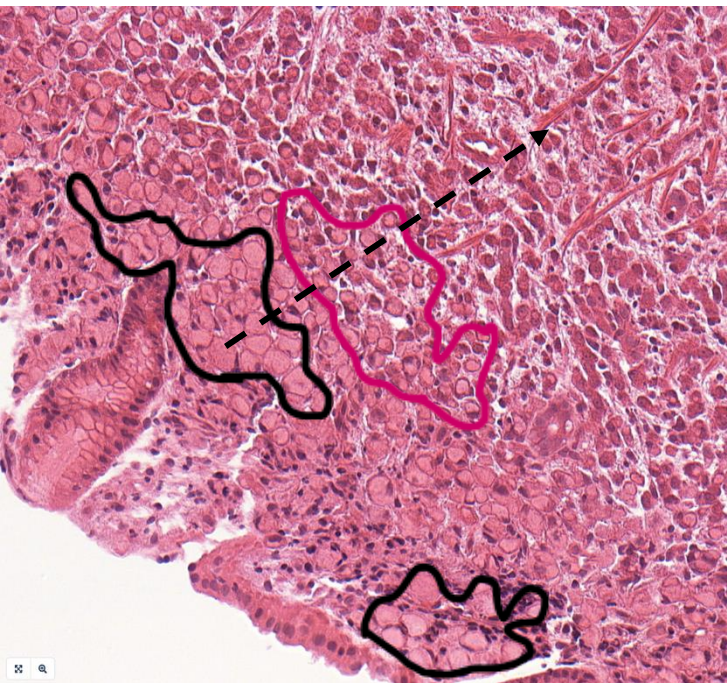
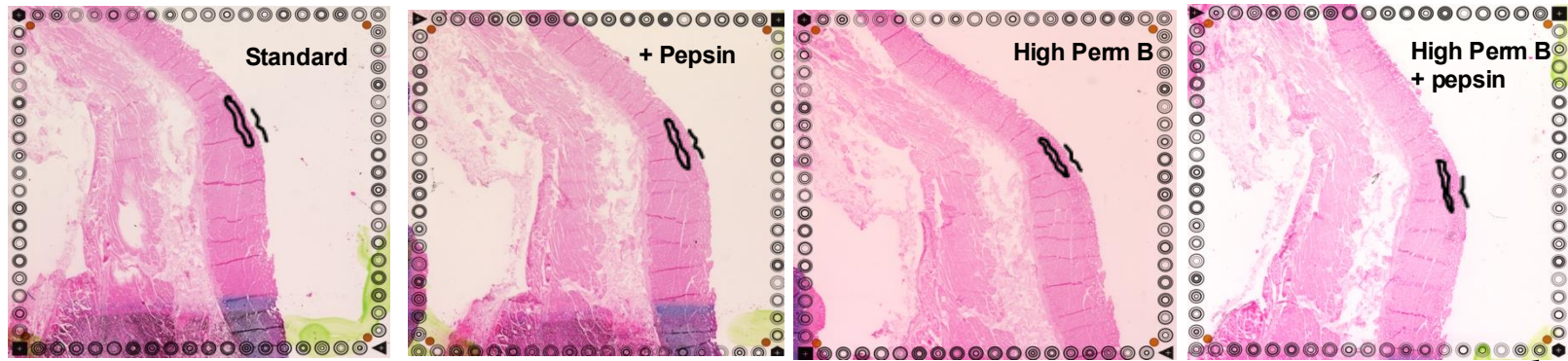
Some tissue types might still present with more challenges. It is important to understand your tissue type and optimize the protocol accordingly

Signet ring cells (SRCs) in diffuse gastric cancer in patients with CDH1 germline mutation



(Monster et al. Biochimica et Biophysica Acta -Reviews on Cancer, 2022)

Major loss of "indolent" mucin filled SRC regions post Visium HD QC



Visium HD Protocol Optimization to release the possibility of mucin trapping the transcript release:

1. Normal Processing
2. +Pepsin
3. High PermB
4. High PermB + Pepsin

Visium HD FFPE Tissue Preparation Handbook 2.0

Handbook | Visium HD FFPE Tissue Preparation 2.0

CG001676 | Rev A

- c. Prepare Decrosslinking Mix according to the appropriate table below. Add reagents in the order listed. Pipette mix thoroughly. Centrifuge briefly. Maintain at room temperature.

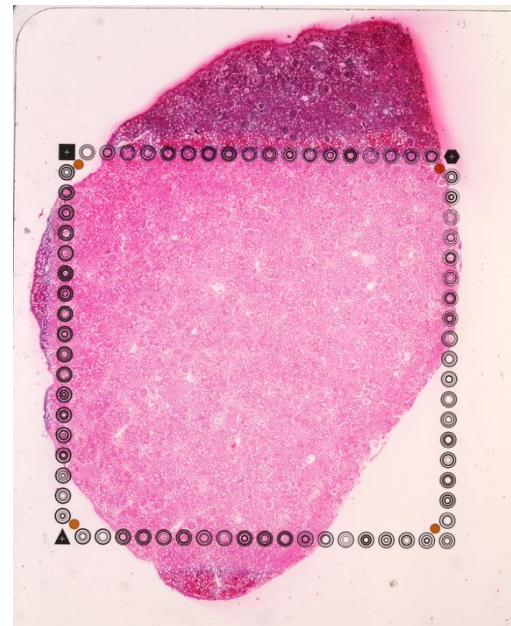
6.5 mm

6.5 mm Gaskets							
	Decrosslinking Mix 1 slide = 1 Tissue Slide	Stock	Final	10x PN	1 slide (µl)	2 slides + 10% (µl)	4 slides + 10% (µl)
●	Decrosslinking Buffer B	-	-	2001094	92.50	203.50	407.00
	Urea	8 M	0.5 M	-	6.25	13.75	27.50
	Diluted Perm Enzyme B (prepared in step b)	-	-	-	1.25	2.75	5.50
	Total	-	-	-	100	220	440

	Protocol 2.0	Previous Protocol Version ("1.0")
Key Updated Steps	• New FFPE tissue QC metric: RNA quantity per cell total RNA per cell • FFPE tissue preparation includes updated decrosslinking condition • Fresh frozen and fixed frozen tissue preparation includes an isopropanol wash • Elution step for all tissue types changes temperature	N/A
Assay Compatibility	Visium HD WTP 6.5 mm and 11 mm	Visium HD WTP 6.5 mm
Timeline	Launching April 2026	Archiving August 2026

1. **Consider starting with a pilot on serial sections to test whether the V1 vs V2 Protocol** with the added **concentration of Perm B enzyme** works best for your tissue of choice
2. Check for the **sequencing saturation**, with high concentration of Perm B, you might need to opt for higher reads/bin
3. If your tissue of choice is not prone to detachment, consider running a plot experiment with serial sections using **Standard Glass Slide versus the Hydrogel-coated Slide**

Live Demo Loupe Browser 9 Manual alignment and json file extraction



Visium HD Spatial Gene Expression Library, Human Kidney (FFPE) - 10X genomics

<https://www.10xgenomics.com/datasets/visium-hd-cytassist-gene-expression-libraries-human-kidney-ffpe>